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Research Interests

3D Electron Microscopy (EM) Organelle Segmentation and Analysis: Instance segmentation and representation learning for mitochondria, nuclei, and pan-organelle structures in volume EM data.

Label-Efficient 3D Medical Image Segmentation: Semi-/weakly-supervised learning with consistency regularization, contrastive learning, and self-training for robust cross-domain generalization.

Structure-Aware and Multi-Resolution Modeling: Integrating skeleton/geometry/topology priors with cross-resolution context modeling and mixture-of-experts strategies for complex 3D segmentation tasks.

Open Benchmarks and Reproducible Evaluation: Building datasets and benchmarks (e.g., MitoEM 2.0, CellMap-related tasks) to support fair and scalable method comparison.

Education

2019 – 2024	Shanghai Jiao Tong University Ph.D. in Biomedical Engineering	Shanghai, China
	• Advisor: Guoyan Zheng • Dissertation: Semi-Supervised Medical Image Analysis Based on Consistency Regularization	
2015 – 2018	University of Chinese Academy of Sciences M.Sc. in Signal and Information Processing	Beijing, China
	• Ranked 1st in the postgraduate entrance examination • Thesis: Object Detection in High-Resolution Remote Sensing Images Based on Deep Learning	
2011 – 2015	China University of Geosciences B.Sc. in Geographic Information System	Beijing, China
	• GPA: 3.62/4.0 • Ranking: 8/60	

Employment History

Aug 2024 – Present	Boston College, Department of Computer Science Postdoctoral Researcher.	Boston, MA, USA
	• Advisor: Prof. Donglai Wei • Leading research on 3D EM segmentation for mitochondria, nuclei, and pan-organelle analysis. • Built and advanced benchmark-driven projects including MitoEM 2.0 (under review at <i>Scientific Data</i>). • Won 1st place in the CellMap Segmentation Challenge (2026; eFIB-SEM, 40+ organelle classes).	
Jul 2017 – Jul 2019	Meituan Search and Recommendation Algorithm Engineer.	Beijing, China
	• Developed ranking and recommendation models for the travel business to improve CTR and CVR. • Methods: Graph Embedding, DNN, and Wide&Deep architectures.	

Research Publications(* indicates equal contribution)

Selected Publications

- 1 **P. Liu** and G. Zheng, "Handling imbalanced data: Uncertainty-guided virtual adversarial training with batch nuclear-norm optimization for semi-supervised medical image classification," *IEEE Journal of Biomedical and Health Informatics*, vol. 26, no. 7, pp. 2983–2994, 2022, (JCR Q1, IF=6.8).
- 2 **P. Liu** and G. Zheng, "Cvcl: Context-aware voxel-wise contrastive learning for label-efficient multi-organ segmentation," *Computers in Biology and Medicine*, vol. 160, p. 106 995, 2023, (JCR Q1, IF=6.3).
- 3 **P. Liu** and G. Zheng, "Context-aware voxel-wise contrastive learning for label efficient multi-organ segmentation," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2022, pp. 653–662, (EI, CCF B).
- 4 **P. Liu** and G. Zheng, "Cr-gloco: Cross-resolution learning via global-local context consistency for semi-supervised 3d medical segmentation," *Computerized Medical Imaging and Graphics*, 2026, Accepted and available online (in press), (JCR Q1, IF=4.9).
- 5 **P. Liu**, B. Shen, L. Liu, Q. Wang, S. Zhang, A. Bhardwaj, I. Arganda-Carreras, K. Narayan, and D. Wei, "Mitoem 2.0: A benchmark for challenging 3d mitochondria instance segmentation from em images," Under Review at *Scientific Data*, 2025, (JCR Q1, IF=6.9).

Journal Articles

- 1 C. Chen*, **P. Liu***, and Y. Song, "Diagnostic performance for severity grading of hip osteoarthritis and osteonecrosis of femoral head on radiographs: Deep learning model vs. board-certified orthopaedic surgeons," *Osteoarthritis Imaging*, vol. 3, no. 2, p. 100 092, 2023.
- 2 **P. Liu**, M. Zhang, X. Gao, B. Li, and G. Zheng, "Joint lymphoma lesion segmentation and prognosis prediction from baseline fdg-pet images via multitask convolutional neural networks," *IEEE Access*, vol. 10, pp. 81 612–81 623, 2022, (JCR Q2, IF=3.6).
- 3 Z. Li, K. Chen, **P. Liu**, X. Chen, and G. Zheng, "Entropy and distance maps-guided segmentation of articular cartilage: Data from the osteoarthritis initiative," *International Journal of Computer Assisted Radiology and Surgery*, pp. 1–8, 2022, (JCR Q2).

Conference Proceedings

- 1 **P. Liu** and G. Zheng, "Semi-supervised learning regularized by adversarial perturbation and diversity maximization," in *Machine Learning in Medical Imaging: 12th International Workshop, MLMI 2021, Held in Conjunction with MICCAI 2021*, Springer, 2021, pp. 199–208, (EI).
- 2 Q. Zhou*, **P. Liu***, and G. Zheng, "Context-aware cutmix is all you need for universal organ and cancer segmentation," in *Fast, Low-Resource, and Accurate Organ and Pan-Cancer Segmentation in Abdomen CT*, Springer, 2023, (EI).
- 3 Q. Zhou, **P. Liu**, and G. Zheng, "Partially supervised multi-organ segmentation via affinity-aware consistency learning and cross site feature alignment," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2023, pp. 671–680.

Under Review

- 1 **P. Liu**, B. Shen, L. Liu, Q. Wang, S. Zhang, A. Bhardwaj, I. Arganda-Carreras, K. Narayan, and D. Wei, "Mitoem 2.0: A benchmark for challenging 3d mitochondria instance segmentation from em images," Under Review at Scientific Data, 2025, (JCR Q1, IF=6.9).

In Preparation

- 1 **P. Liu** and D. Wei, *Skelaaffinity-seg: Skeleton-guided geometry and global affinity learning for complex 3d tubular instance segmentation*, To be submitted to IEEE Transactions on Medical Imaging, 2026.
- 2 **P. Liu** and D. Wei, *Mixture-of-experts for multi-resolution pan-organelle segmentation in 3d volume electron microscopy*, To be submitted to IEEE Transactions on Medical Imaging, 2026.

Competitions and Scholarships

Competitions

- 2026 **1st Place**, CellMap Segmentation Challenge (eFIB-SEM multi-organelle segmentation, 40+ classes).
- 2023 **2nd Place**, MICCAI FLARE 2023 Challenge. In MICCAI 2023 workshop.
- 2018 **Ranking: 20/120**, MDD Cup 2018 of Meituan Group. In Kaggle MDD Cup 2018.
- 2017 **Ranking: 7/148**, MDD Cup 2017 of Meituan Group. In Kaggle MDD Cup 2017.

Scholarships

- 2014 **Third-class Scholarship**. Awarded by China University of Geosciences.
- 2013 **Third-class Scholarship**. Awarded by China University of Geosciences.
- 2012 **National Encouragement Scholarship**. Awarded by Ministry of Education of China.
 Second-class Scholarship. Awarded by China University of Geosciences.

Teaching Assistant

Shanghai Jiao Tong University, CS 2901: Programming Ideas and Methods (C), Fall 2021 and Fall 2022.
Shanghai Jiao Tong University, CS 170: Programming Ideas and Methods (C), Fall 2019 and Fall 2020.

Courses I Can Teach

Programming Fundamentals (C/C++), Data Structures and Algorithms
Biomedical Image Processing and Analysis

Peer Reviewer

IEEE Transactions on Medical Imaging
Medical Image Analysis
IEEE Journal of Biomedical and Health Informatics
Neural Networks
Computerized Medical Imaging and Graphics
MICCAI 2022
MICCAI 2021

Skills

Languages	English (professional working proficiency; TOEFL, CET-6), Mandarin Chinese (native).
Coding	Python, C++, PyTorch, Linux, Git, scientific computing workflows for large-scale 3D imaging data.